

Summary of Malabsorption Types and Diagnostics

Malabsorption Type	Primary Nutrient Affected	Key Symptom/Sign	Diagnostic Tests	Treatment
Fat Malabsorption	Fat	Steatorrhea ($\geq 20\text{g}$ fat/day)	72-hour stool collection for fecal fat	Address underlying cause; fat-soluble vitamin supplementation.
Protein Malabsorption	Protein	Edema, muscle atrophy, hypoalbuminemia	Serum albumin/protein levels	Address underlying cause; nutritional support.
Carbohydrate Malabsorption	Carbohydrates	Watery diarrhea, flatulence, bloating	D-Xylose test (serum ≤ 20 mg/dL, urine < 4.5 g/5hr), Hydrogen breath test	Dietary restriction of offending carbohydrate.
Global Malabsorption	All Nutrients	Chronic diarrhea, steatorrhea, weight loss	Combination of tests depending on suspected cause.	Treatment of underlying cause.
Celiac Disease	Global (primarily)	Chronic diarrhea, steatorrhea, anemia, bloating	TTGA IgA, Upper endoscopy with duodenal biopsy (villous atrophy, crypt hyperplasia, increased IELs), HLA-DQ2/DQ8 testing	Strict gluten-free diet.
Tropical Sprue	Global	Chronic diarrhea, steatorrhea, weight loss	Diagnosis of exclusion; rule out infections. Small intestinal biopsy may show non-specific changes.	Tetracycline, Folic acid.

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Exocrine Pancreatic Insufficiency	All Nutrients	Steatorrhea, weight loss, abdominal pain	Fecal elastase-1 < 200 $\mu\text{g/g}$, Serum trypsinogen < 20 ng/mL, ERCP	Pancreatic enzyme replacement therapy.
Whipple's Disease	Global	Diarrhea, steatorrhea, weight loss, arthralgias, neurological symptoms	PAS stain of biopsy, PCR for <i>T. whipplei</i>	IV antibiotics followed by oral maintenance therapy.
Lactose Intolerance	Lactose	Bloating, diarrhea after dairy intake	Hydrogen breath test ($\uparrow$ hydrogen), Stool pH ($\downarrow$), Stool osmolality ($\uparrow$), Oral lactose challenge	Lactose-free diet, lactase supplements.

Ulcerative Colitis (UC) vs. Crohn's Disease (CD) Comparison

Feature	Ulcerative Colitis	Crohn's Disease
Age Group	Any	Any
Gender	Male = Female (M=F)	Slight female preponderance
Incidence	Stable	Increasing
Ethnic Group	Any	Any; more common in Ashkenazi Jews
Genetic Factors	HLA-DR*103, colonic epithelial barrier function (HNF4a, LAMB1, CDH1)	Defective innate immunity and autophagy (NOD2, ATG16L1, IRGM)
Risk Factors	More common in non-/ex-smokers; Appendicectomy protects	More common in smokers
Anatomical Distribution	Colon only; begins at anorectal margin with variable proximal extension	Any part of gastrointestinal tract; perianal disease common; patchy distribution, skip lesions
Extra-intestinal Manifestations	Common	Common
Presentation	Bloody diarrhoea	Variable; pain, diarrhoea, weight loss all common
Histology	Inflammation limited to mucosa; crypt distortion, cryptitis, crypt abscesses, loss of goblet cells	Submucosal or transmural inflammation common; deep fissuring ulcers, fistulae; patchy changes; granulomas
Management	5-ASA; glucocorticoids; azathioprine; biological therapy (anti-TNF, anti- $\alpha_4\beta_7$ integrin); colectomy is curative	Glucocorticoids; azathioprine; methotrexate; biological therapy (anti-TNF, anti- $\alpha_4\beta_7$ integrin); nutritional therapy; smoking cessation; surgery for complications is not curative; 5-ASA is not effective